

BUSINESS INTELLIGENCE REPORT

Car Sales Data Analysis and Dashboard Development

Week 3 Task 03 — Business Intelligence & Dashboard Development

Data Science Internship — Devixo Solutions

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Executive Summary

This report presents a business intelligence analysis of a 50,000-record car sales dataset covering five manufacturers (Ford, Volkswagen, Toyota, BMW, and Porsche) and vehicles manufactured between 1984 and 2022. The analysis covers data cleaning and validation, feature engineering, key performance indicator calculation, exploratory data analysis, and a supporting predictive pricing model.

The dataset was found to be complete, with no missing values or duplicate records across all 50,000 rows. Ford and Volkswagen make up the largest share of vehicles by volume, while Porsche and BMW command the highest average prices despite far lower volume. Vehicle age and mileage both show a negative relationship with price, with age being the stronger of the two factors. A Random Forest regression model was built to predict vehicle price and achieved a high accuracy level ($R^2 = 0.998$), confirming that manufacturer, model, engine size, fuel type, year of manufacture, and mileage together explain most of the variation in price.

The report concludes with 15 business insights and 10 business recommendations intended to support pricing, inventory, and stocking decisions.

1. Introduction and Objective

The objective of this task was to transform raw car sales data into actionable business insights by cleaning the data, engineering relevant features, calculating key performance indicators, building a supporting dashboard of visualizations, and producing business recommendations. The dataset used is a car sales dataset of 50,000 records with seven original fields: Manufacturer, Model, Engine size, Fuel type, Year of manufacture, Mileage, and Price.

2. Data Overview

The dataset contains 50,000 rows and 7 original columns. Manufacturers present are Ford, Volkswagen, Toyota, BMW, and Porsche, and fuel types are Petrol, Diesel, and Hybrid. Vehicles range from year of manufacture 1984 to 2022, engine size 1.0L to 5.0L, mileage 630 to 453,537, and price \$76 to \$168,081.

3. Data Cleaning and Validation

The dataset was checked for data types, missing values, duplicate records, invalid numerical values, and categorical consistency.

- No missing values were found in any of the 50,000 rows.
- No duplicate records were identified.
- Numerical fields (engine size, mileage, year of manufacture, price) were checked for invalid or non-positive values; none were found.
- Categorical fields (manufacturer, fuel type) were checked for inconsistent naming and formatting; none were found.
- Outliers in price (2,649 records, 5.3%) and mileage (385 records, 0.8%) were investigated using the IQR method and retained, as they represented valid high-value or high-mileage vehicles rather than data errors.

4. Feature Engineering

Two derived features were created to support deeper analysis:

- Vehicle Age — calculated as the reference year (2026) minus year of manufacture.
- Mileage per Year — mileage divided by vehicle age, indicating usage intensity.
- Price per Mileage — price divided by mileage, indicating value retained per mile driven.

5. Key Performance Indicators

The following KPIs summarize the overall dataset:

Key Performance Indicator	Value
Total Vehicles Listed	50,000
Total Portfolio Value	\$691,445,158
Average Price	\$13,828.90
Median Price	\$7,971.50
Average Mileage	112,497 miles
Average Vehicle Age	21.8 years
Average Engine Size	1.77 L

6. Dashboard: Visual Analysis

The visualizations below are taken directly from the notebook's own analysis output and summarize vehicle volume, pricing, and the effect of mileage and age on price.

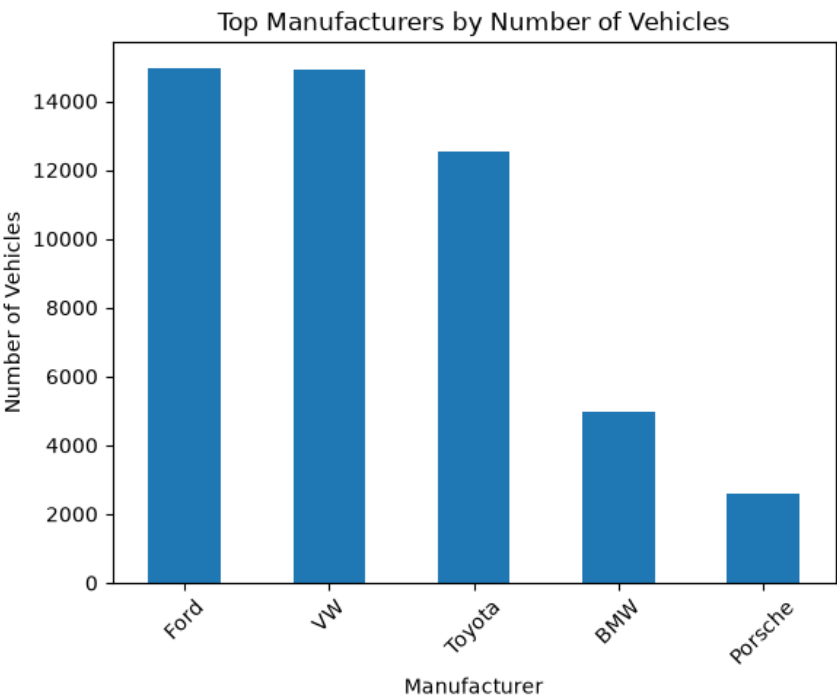


Figure 1: Top Manufacturers by Number of Vehicles

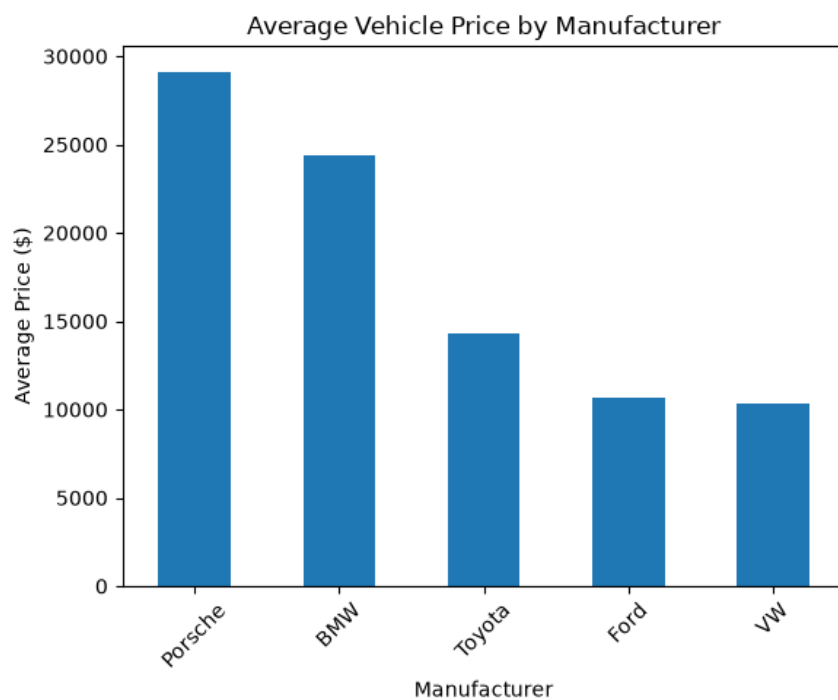


Figure 2: Average Vehicle Price by Manufacturer

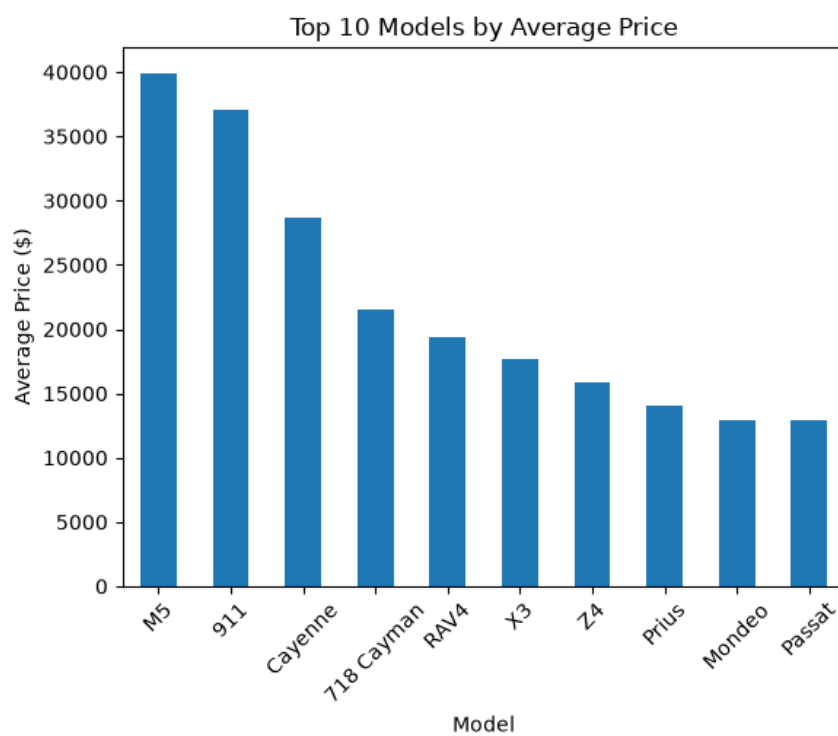


Figure 3: Top 10 Models by Average Price

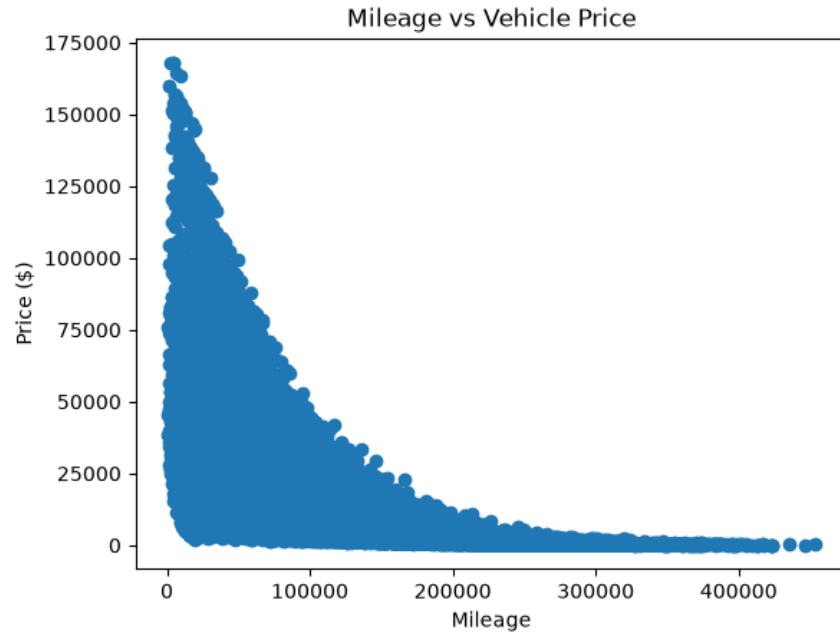


Figure 4: Mileage vs Vehicle Price

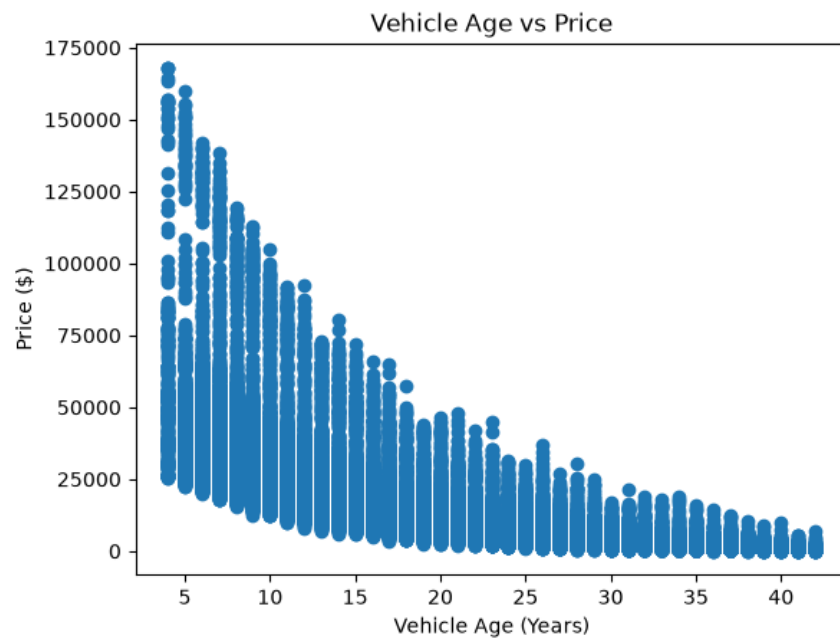


Figure 5: Vehicle Age vs Price

7. Exploratory Data Analysis Findings

7.1 Manufacturer and Model Analysis

Ford (14,956 vehicles) and Volkswagen (14,906 vehicles) together account for roughly 60% of the portfolio, followed by Toyota (12,553), BMW (4,964), and Porsche (2,609). Despite the lowest volume,

Porsche has the highest average price (\$29,103.76), followed by BMW (\$24,431.68). Ford and Volkswagen have the lowest average prices, both close to \$10,500.

At the model level, the BMW M5 has the highest average price (\$39,911.11), followed by the Porsche 911 (\$37,024.72) and the Porsche Cayenne (\$28,714.57). The single most expensive listings in the dataset are individual BMW M5 and Porsche 911 units, several priced above \$160,000.

7.2 Price Relationships

Mileage shows a moderate negative correlation with price ($r = -0.63$): higher-mileage vehicles are consistently priced lower. Vehicle age shows a stronger negative correlation with price ($r = -0.71$), indicating that age is a more significant driver of depreciation than mileage alone.

7.3 Fuel Type and Engine Size

Petrol vehicles make up the largest share of the portfolio (25,488 vehicles, 51%), followed by Diesel (13,268, 27%) and Hybrid (11,244, 22%). The most common engine size is 1.4L (10,060 vehicles), followed by 2.0L (7,631) and 1.8L (7,032), indicating that the portfolio leans toward smaller, economy-oriented engines.

8. Predictive Pricing Model (Bonus)

A Random Forest Regressor was trained using manufacturer, model, engine size, fuel type, year of manufacture, and mileage as predictors of price, with an 80/20 train-test split.

Evaluation Metric	Value
Mean Absolute Error (MAE)	\$290.28
Root Mean Squared Error (RMSE)	\$644.53
R ² Score	0.9985

The model's high accuracy ($R^2 = 0.998$) confirms that the available fields together are strong predictors of vehicle price, and the model can support internal price estimation or listing validation.

9. Business Insights

1. Ford and Volkswagen dominate the inventory, together accounting for about 60% of all 50,000 listed vehicles.
2. Porsche is the smallest segment by volume (2,609 vehicles, 5.2%) but commands the highest average price of any manufacturer.
3. The average vehicle price is \$13,828.90, while the median is \$7,971.50 — the gap indicates a right-skewed price distribution driven by a smaller number of premium vehicles.
4. Porsche vehicles average \$29,103.76 and BMW vehicles average \$24,431.68 — both roughly double or more the Ford and Volkswagen averages (around \$10,500).
5. The BMW M5 and Porsche 911 are the two most expensive individual models in the dataset, with several listings exceeding \$160,000.
6. Across all models, the M5 has the highest average price (\$39,911.11), followed by the 911 (\$37,024.72) and the Cayenne (\$28,714.57).
7. Vehicle mileage and price show a moderate negative correlation ($r = -0.63$): higher mileage is consistently associated with lower price.

8. Vehicle age shows a stronger negative correlation with price ($r = -0.71$), making age a more influential depreciation factor than mileage.
9. The average vehicle age is 21.8 years, meaning most of the portfolio consists of older, already-depreciated vehicles rather than nearly new stock.
10. Petrol vehicles are the most common fuel type (51%), followed by Diesel (27%) and Hybrid (22%).
11. The most common engine size is 1.4L (10,060 vehicles), suggesting the portfolio leans toward smaller, economy-oriented vehicles.
12. Price outliers (2,649 records, 5.3%) are concentrated among premium brands and represent legitimate high-value vehicles rather than data errors.
13. Mileage outliers are far fewer (385 records, 0.8%), indicating mileage data is comparatively clean and consistent.
14. No missing values or duplicate records were found across the full 50,000-row dataset, indicating high overall data quality.
15. A Random Forest regression model predicted vehicle price with high accuracy ($R^2 = 0.998$, $MAE \approx \$290$), confirming that manufacturer, model, engine size, fuel type, year, and mileage together strongly determine price.

10. Business Recommendations

1. Prioritize marketing and stock allocation toward Ford and Volkswagen, which represent the bulk of transaction volume.
2. Selectively expand premium-segment sourcing (Porsche, BMW), since these brands generate disproportionate value per unit despite low volume.
3. Use vehicle age, more than mileage, as the primary factor when setting or negotiating listing prices.
4. Flag listings with unusually high mileage for price review, since mileage still meaningfully reduces value.
5. Retain legitimate high-value outliers (e.g., low-mileage M5 or 911 units) in pricing models rather than removing them as data errors.
6. Grow the Hybrid vehicle segment, currently the smallest fuel-type category, to capture rising demand for fuel-efficient vehicles.
7. Use the trained price-prediction model ($R^2 = 0.998$) as an internal valuation tool to standardize and validate pricing decisions.
8. Introduce sale date, region, and customer identifier fields going forward to enable trend, regional, and customer-level analysis not possible with the current dataset.
9. Monitor the 1.4L-and-below economy engine segment closely, as it forms the largest single band and likely drives high-turnover inventory.
10. Apply differentiated pricing strategies by manufacturer tier — economy (Ford, Volkswagen, Toyota) versus premium (BMW, Porsche) — rather than a single blanket pricing approach.

11. Data Limitations

The dataset does not include sale date, customer, or region fields. As a result, monthly revenue trends, top-customer analysis, and regional performance — items referenced in the standard dashboard template —

could not be produced from this data. The KPIs and visualizations in this report are limited to manufacturer, model, price, mileage, engine size, and fuel-type dimensions that are actually present in the dataset.

12. Conclusion

This analysis shows a car sales portfolio led by Ford and Volkswagen in volume and Porsche and BMW in average price, with vehicle age as the strongest driver of price depreciation. Data quality was high, with no missing values or duplicates, and a supporting Random Forest model confirmed that the available fields can predict price with high accuracy. The insights and recommendations above provide a basis for pricing strategy, inventory prioritization, and future data collection improvements.